CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 13: WRITE THE PYTHON POGRAM TO IMPLEMENT MINIMAX ALGORITHM FOR GAMING

AIM

To write a Python program to implement the Minimax Algorithm for gaming using the Tic-Tac-Toe game.

ALGORITHM

1. Start the program.
2. Create a Tic-Tac-Toe board.
3. Check whether the current board represents a win, loss, or draw.
4. If the game is over, return the corresponding score.
5. Generate all possible moves for the current player.
6. If it is the maximizing player's turn, choose the move with the maximum score.
7. If it is the minimizing player's turn, choose the move with the minimum score.
8. Repeat until the best move is found.
9. Display the best move and its score.
10. Stop the program.

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```
            if board[i] == " ":
                board[i] = "O"
                score = minimax(board, True)
                board[i] = " "
                best_score = min(score, best_score)

        return best_score


def check_win(board, player):
    wins = [
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)
    ]

    for a, b, c in wins:
        if board[a] == board[b] == board[c] == player:
            return True

    return False


def find_best_move(board):
    best_score = -float("inf")
    best_move = -1

    for i in range(9):
        if board[i] == " ":
            board[i] = "X"
            score = minimax(board, False)
            board[i] = " "

            if score > best_score:
                best_score = score
                best_move = i

    return best_move, best_score


board = [" "] * 9

move, score = find_best_move(board)

print("Tic-Tac-Toe using Minimax Algorithm")
print("Best Move Position:", move + 1)
print("Best Score:", score)
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/navit/PY8.py ==========
Tic-Tac-Toe using Minimax Algorithm
Best Move Position: 1
Best Score: 0
>>>
```

RESULT

The Python program was executed successfully, and the Minimax algorithm determined the best move for the maximizing player (X) in Tic-Tac-Toe.